Project – Presentation (Mon 4/20 and Wed 4/22)

- A 20 minute presentation that will be presented to the class. The presentation should minimally cover:
  - Overview of the project, AWS technologies used
  - A <u>working</u> demo running on AWS with a <u>polished interface</u>
  - No setup or teardown needed for demo
  - Presentation (minimally 8-10 slides) should be in PPT or PDF format
    - Present a comprehensive project overview including AWS technologies used, architecture, observability approach, and cost analysis
    - Conclude with reflections on lessons learned, cloud benefits, challenges, AI tool usage, and potential future enhancements
  - Details on myCourses at Content > Term Project > 614 Term Project
  - <u>Everyone</u> presents
- **<u>Note:</u>** Teams that exceed the 20-minute time limit will lose points
  - It is essential to rehearse your presentation in advance to ensure it stays within the allotted time

Project - User's guide (Due 4/27)

- A user's guide should be submitted that includes:
  - Full description on how to **setup and use** your project
  - User's guide can be submitted either as a Word/PDF Document, md (Markdown) or HTML
  - Include examples, where relevant
  - There is no minimal length required

Project - Submission Deliverables

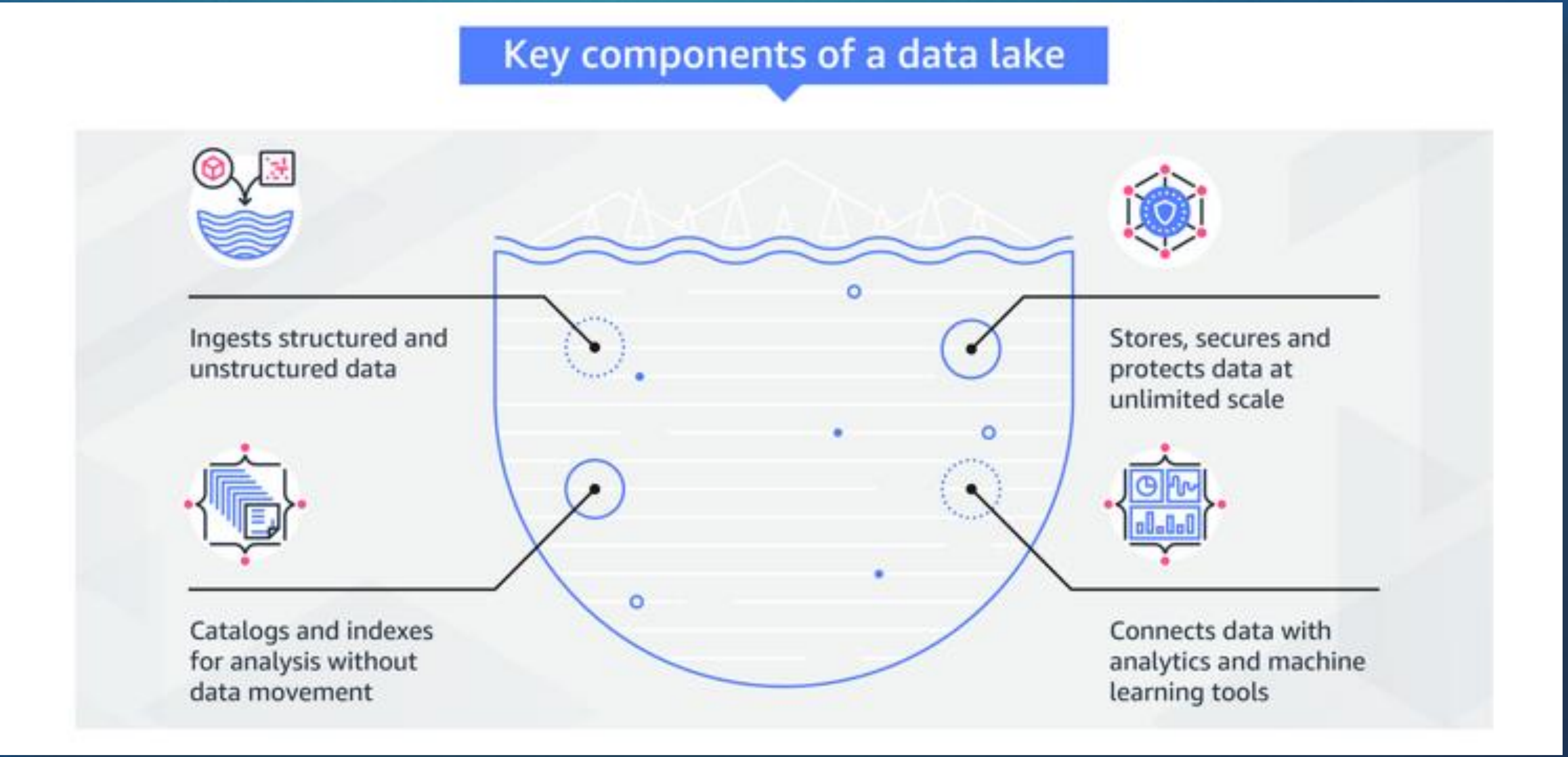
- For each team, presentation, white paper and user's guide should be uploaded Assignments > Term Project > Project Submission
  - All code developed for this project, sample data (if any) and other relevant files must be uploaded to Github Classroom
  - For the presentations (proposal and final) and whitepaper, we will be using Google Docs so team progress and participation can be tracked
- Grading Rubric:
  - Contains details for each deliverable of the project
  - See Content > Term Project > Term Project Grade Rubric

RECAP

- Big Data Analytics is the complex process of examining large and varied data sets, or big data, to uncover information such as hidden patterns, unknown correlations, market trends and customer preferences that can help organizations make informed business decisions
- Why important?
  - Cost Reduction
  - New products and services
  - Faster and smarter better decision
  - Time to market reductions

RECAP

- A <u>Data Lake</u> allows you to store all your structured and unstructured data, in one centralized repository, and at any scale
- With a Data Lake, you store your data <u>as-is</u>, without having to first structure the data, based on potential questions you may have in the future



RECAP

- Spark SQL is a Spark module for structured data processing
- It provides a programming abstraction called DataFrames and can also act as a distributed SQL query engine
- It enables unmodified Hadoop Hive queries to run up to 100x faster on existing deployments and data
- It also provides powerful integration with the rest of the Spark ecosystem

Generative AI on the Cloud

Engineering Cloud Software Systems

Department of Software Engineering
Rochester Institute of Technology

Overview

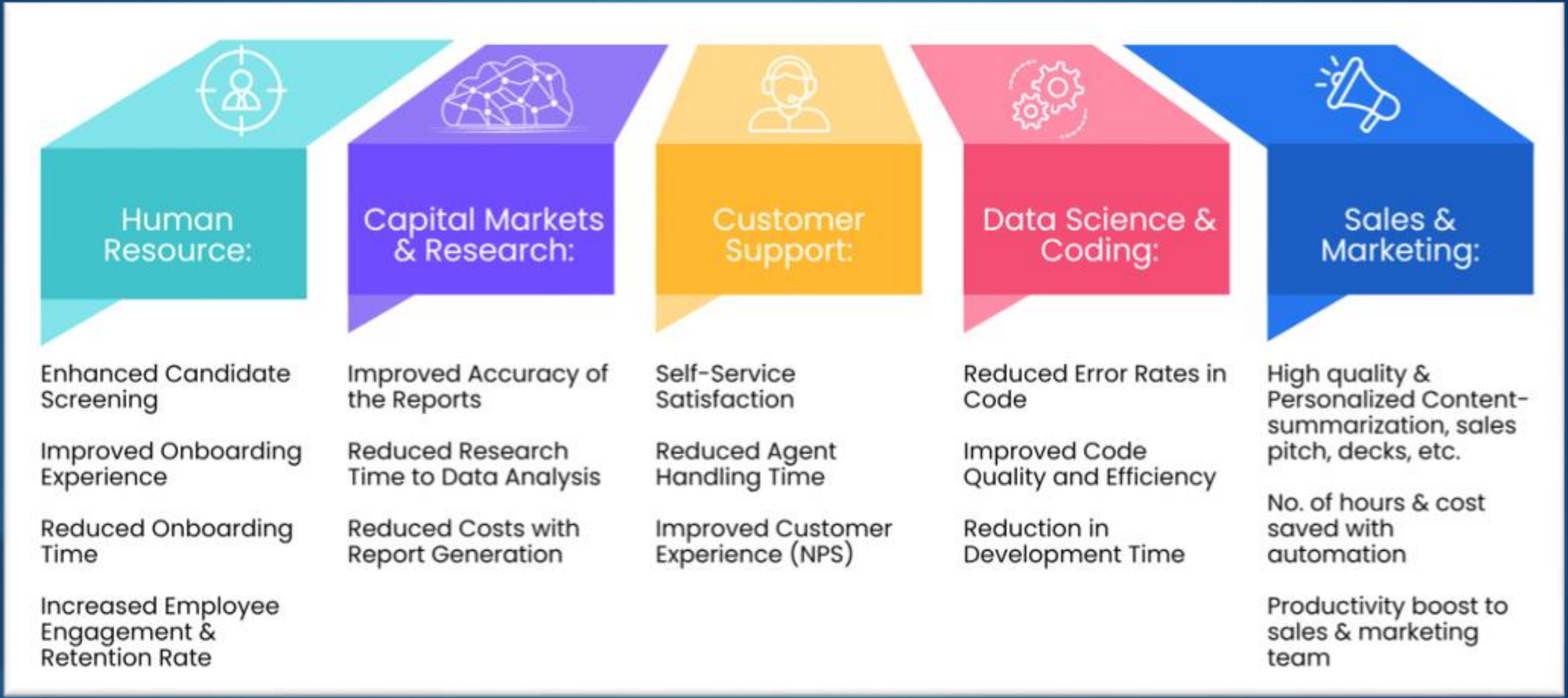
- Generative AI (GenAI) creates new content (text, images, code) rather than only predicting or classifying
- Modern GenAI requires massive compute, data, and orchestration
- Cloud platforms make GenAI accessible through scalable infrastructure and managed services

Source: https://www.idc.com/resource-center/blog/embracing-the-future-how-genai-is-revolutionizing-cloud-infrastructure-for-growing-tech-vendors/

Gen AI vs. Traditional AI

	GenAI	Traditional AI
Purpose and Output	**Specifically designed to create new content**, such as text, images, music, or code, that resembles human-made creations	Used to analyze data, **recognize patterns, make predictions**, or automate tasks without necessarily generating new, creative outputs.
Data Utilization	Leverages large-scale, diverse datasets to develop a **broad understanding** that can be applied to various creative tasks	Models are often trained on more narrowly defined datasets **tailored for specific problem-solving** or decision-making applications.
Interactivity and Adaptability	Models are designed to interact with users in **more creative ways**, producing outputs that can evolve based on user input and context	**More rigid**, focusing on predefined tasks and not necessarily designed for creative adaptability

Common GenAI Use Cases

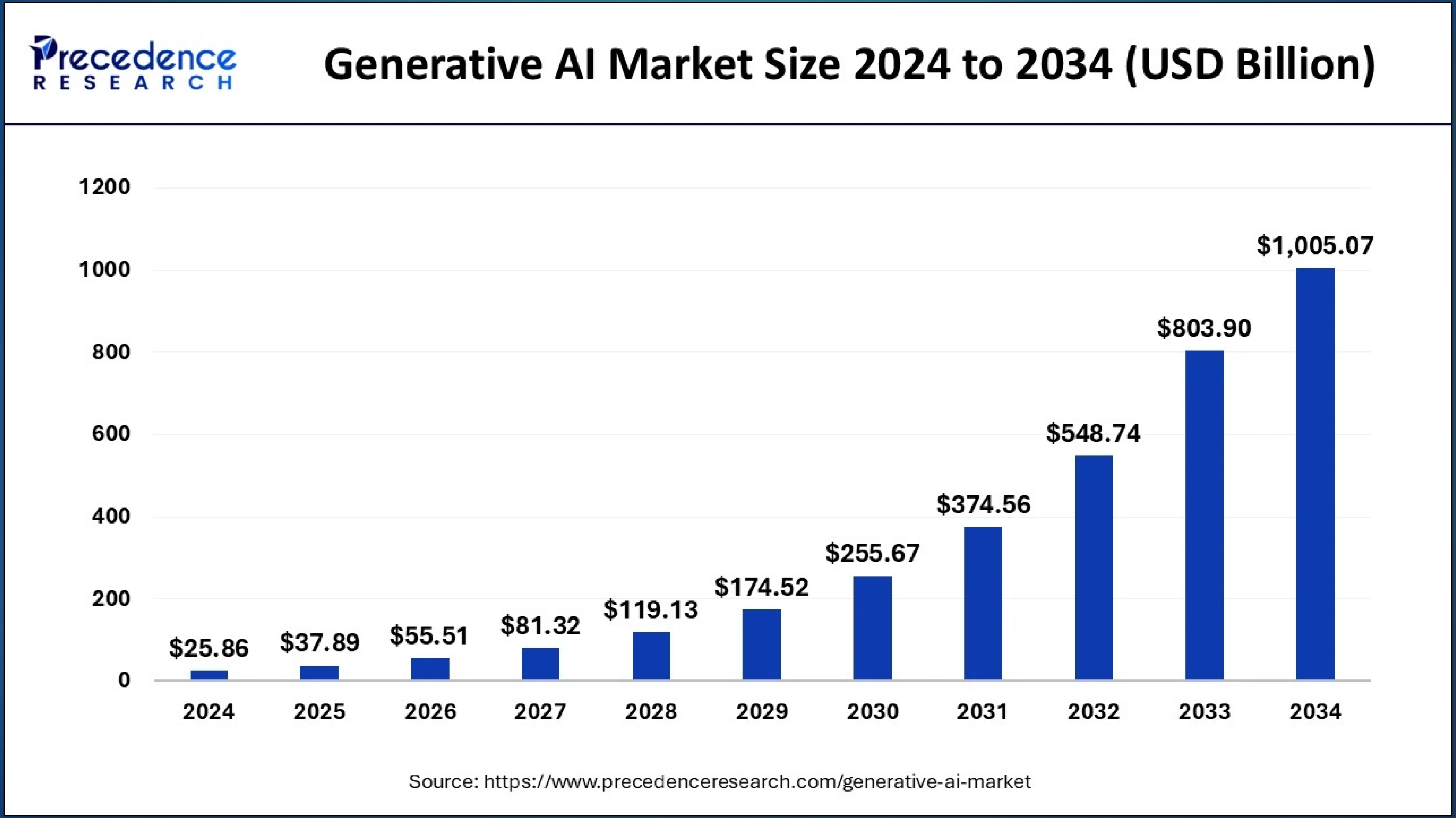


- These use cases are practical only at scale because of cloud compute, APIs, and managed services

Source: https://www.fluid.ai/blog/how-do-you-measure-gen-ai-deployment-pilot-success-key-performance-indicators-and-metrics

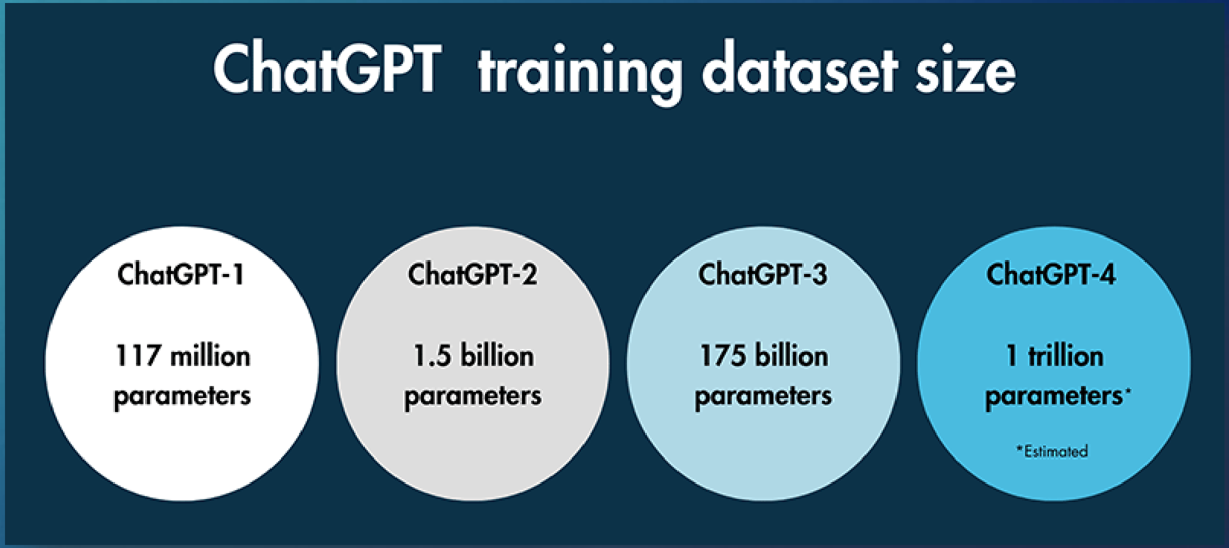
GenAI Market Size and Growth

- Growth is driving cloud providers to invest heavily in managed GenAI platforms



ChatGPT: Scale and Cloud Reality

- ChatGPT runs on hyperscale cloud infrastructure (Microsoft Azure), relying on thousands of high-end GPUs
- Training and operating large language models requires enormous compute, often costing millions of dollars per month
- Models are trained on trillions of parameters, far beyond what on-prem systems can support
- Cloud platforms make this scale elastic, global, and accessible via APIs rather than custom hardware



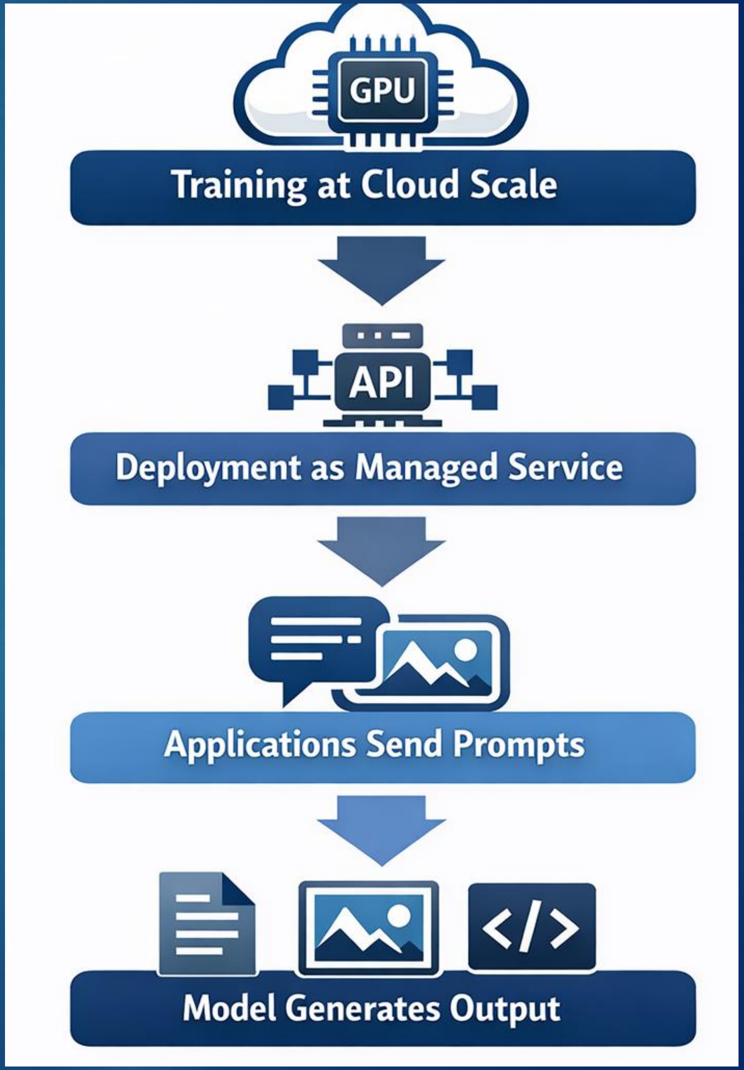
Source: https://www.stylefactoryproductions.com/blog/chatgpt-statistics

Why GenAI Gained Popularity?

- Model Scale Improved → Required Cloud GPUs
  - Larger models (e.g., GPT-3 with 175B parameters) deliver better accuracy and creativity
  - Training these models needs powerful GPUs/TPUs, enabled by cloud platforms
- Cloud Removed Hardware Barriers
  - AWS, Google Cloud, and Azure offer on-demand compute, eliminating costly hardware
  - Broad access to high-performance resources accelerates innovation for all organizations
- APIs Abstracted Complexity
  - Pre-trained models (GPT, DALL·E, Stable Diffusion) available via APIs
  - Developers can integrate AI without deep ML expertise or massive resources

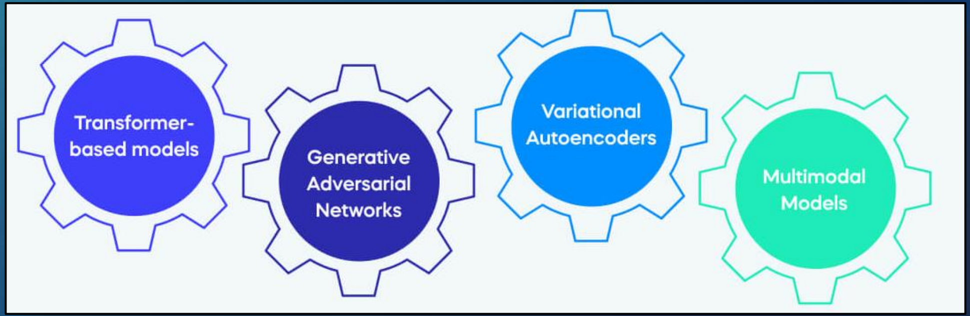
How Does GenAI work?

- Training at Cloud Scale
  - Models trained on massive datasets using GPUs/TPUs in the cloud
  - Done once by AI vendors; costly and resource-intensive
- Deployment as Managed Service
  - Hosted on scalable cloud platforms and exposed via APIs
  - Cloud ensures availability, scaling, and performance
- Applications Send Prompts
  - Apps send text, images, or other inputs via API calls
  - Prompts guide the model, they don't retrain it
- Model Generates Output
  - Predicts and creates new content based on the prompt
  - Outputs include text, images, code, or structured data



Do You Need to Know Model Types?

- Most teams don't build or train models themselves
- Cloud providers deliver pre-trained models via APIs, removing complexity
- Developers focus on:
  - Integrating models into applications
  - Designing prompts and workflows for desired outputs
  - Managing security, cost, and governance
- The cloud handles model infrastructure and scaling, so teams can concentrate on business logic



Source: https://www.jaroeducation.com/blog/future-of-generative-ai/

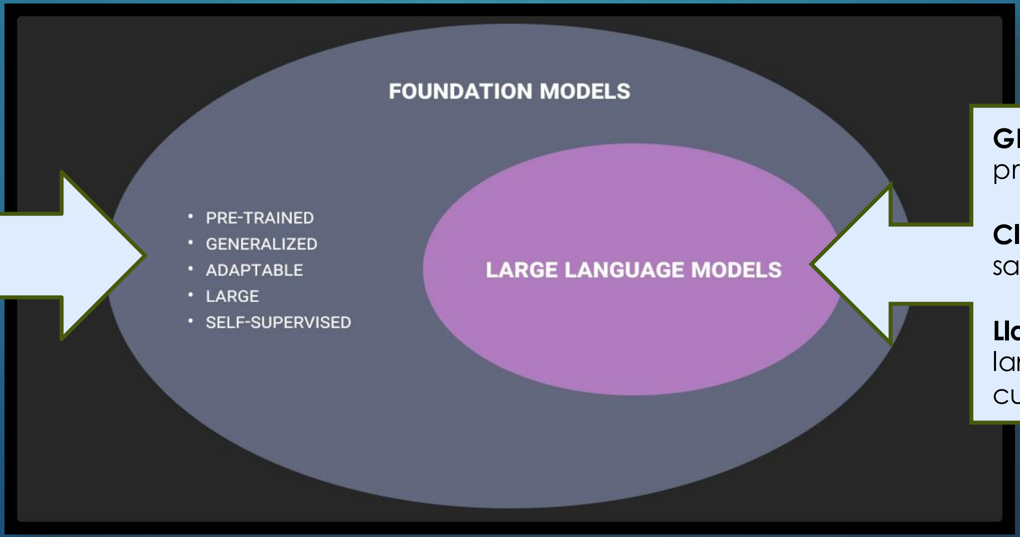
Foundation Models and LLMs

- Foundation Models:
  - Large, pre-trained models that serve as a base for many tasks
  - Adaptable for text, images, and more through fine-tuning or APIs
- Large Language Models (LLMs):
  - Specialized foundation models for text generation, summarization, and code
  - Commonly used for chatbots, assistants, and content creation

CLIP (OpenAI): Connects text and images to understand visual concepts.

Stable Diffusion: A foundational model specialized in image generation.

Segment Anything (Meta): A foundational model for computer vision that can "cut out" any object in any image.



GPT-4 (OpenAI): The engine behind the premium version of ChatGPT.

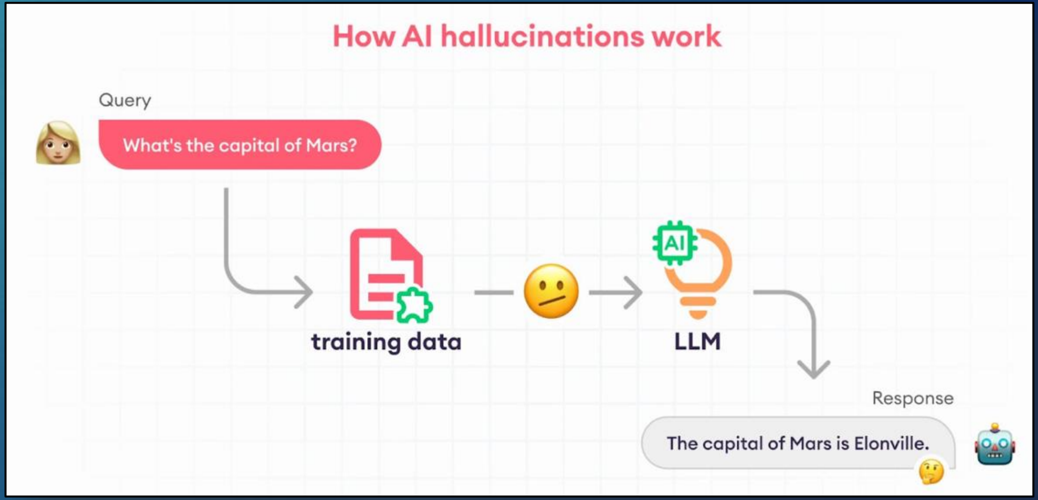
Claude (Anthropic): An LLM focused on safety and long-context conversations.

Llama 3 (Meta): An open-weights large language model used by developers to build custom chatbots.

Source: https://labelyourdata.com/articles/foundation-model-vs-llm

GenAI Hallucinations

- Models can produce plausible but false outputs due to data gaps or conflicts
- Not just a model issue, it's an <u>operational</u> problem
- Risks:
  - Misinformation, reputational harm, compliance failures
- Operation Safeguards:
  - Guardrails & content filters: Block unsafe or misleading outputs before delivery
  - Logging & monitoring: Track prompts and responses to detect patterns and prevent recurrence.
  - Prompt constraints: Enforce structured, safe input to guide models toward accurate results



Source: https://hackernoon.com/gen-ai-hallucinations-the-good-the-bad-and-the-costly

GenAI and Cloud Migration

- Scenario:
  - ACME Company, a major provider of News, Financial, and Legal content, plans a large-scale migration to the cloud. They currently operate entirely on-premises
- Your Role:
  - As a cloud consulting team, you've been hired to propose strategies for achieving this goal
- What if AI helped?
  - AI as an Assistant, Not a Replacement:
  - AI can augment human expertise by analyzing workloads, dependencies, and migration paths
  - It can suggest optimized architectures, cost estimates, and risk mitigation strategies
- Humans remain in control as AI provides insights to accelerate planning and reduce complexity

Source: https://www.computerweekly.com/news/366555296/How-generative-AI-and-cloud-complement-each-other

GenAI and Cloud Migration

Discovery & Planning

Documenting Current Architectures

- Scan environments to construct topological maps of servers, apps, databases
- Generate interactive visualizations and documentation of existing infrastructure

Optimizing Target Architectures

- Analyze dependencies and usage patterns to identify consolidation and modernization opportunities
- Recommend optimized partitioning of systems and data for cloud

Cloud Readiness

Refactoring Codebases

- Migrate legacy code to modern languages better suited for cloud platforms
- Decompose monoliths into distributed microservices applying cloud design patterns

Migrating Data

- Profile legacy databases and extract schemas
- Generate ETL logic to transform and load data into cloud data platforms

Migration & Validation

Automating Deployment

- Generate infrastructure-as-code definitions for provisioning target cloud environments
- Package and deploy applications with their transformed code and data to the cloud

Validating Correctness

- Auto-generate unit, integration and performance tests for migrated apps
- Validate functionality, scalability and compliance with pre-migration configurations

Optimization & Modernization

Incremental Modernization

- Prioritize and schedule incremental modernization of remaining legacy processes
- Leverage AI to convert legacy systems and data incrementally post-migration

Optimizing Cost and Utilization

- Continuously optimize right-sizing of cloud resources based on actual usage data
- Notify on migration elements ripe for modernization to prevent cloud waste

GenAI

Source: https://medium.com/@bijit211987/accelerating-cloud-migration-efficiency-with-generative-ai-fdc7eaea1509

How does the Cloud Support GenAI?

- Elastic Compute
  - On-demand access to GPUs and accelerators for training and inference at scale.
- Massive Data Storage
  - Cloud storage supports the large datasets required for GenAI workloads.
- Cost Flexibility
  - Pay-as-you-go pricing avoids large upfront infrastructure investments.
- Fast Experimentation
  - Resources can be provisioned quickly to test and deploy GenAI solutions
- Managed AI Services
  - Cloud platforms provide pre-trained models, APIs, and tooling that reduce complexity

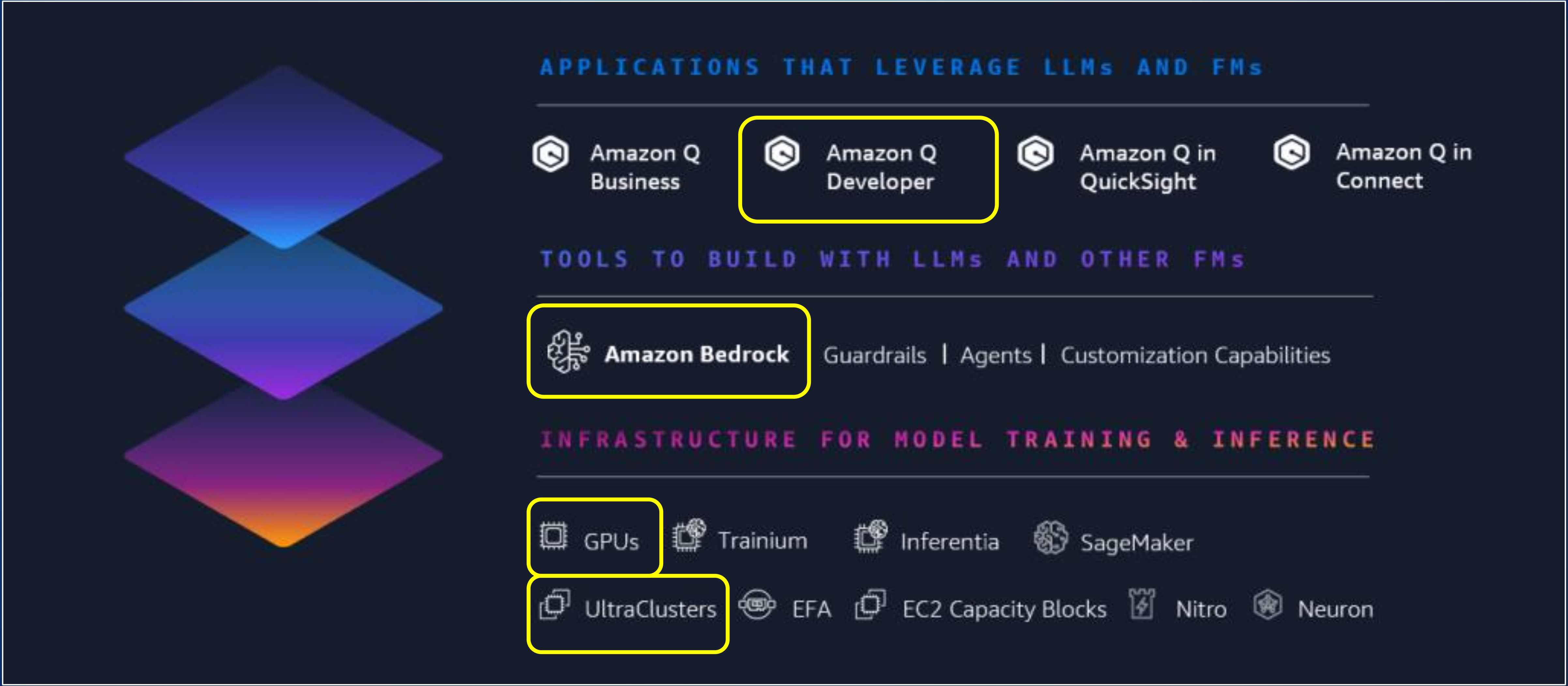
Source: https://blogs.idc.com/2024/07/31/embracing-the-future-how-genai-is-revolutionizing-cloud-infrastructure-for-growing-tech-vendors/

Public Cloud offerings of GenAI Services

GenAI	Comparison of Google, Azure & AWS GenAI Services			
Hyperscaler	**Build & Deploy**	**Foundational Models**	**PrepareData**	**AI Infrastructure**
Google	VertexAI	PaLM (Language)	Data Set	supports NVIDIA L4 and H100 Tensor Core GPUs, as well as prior GPU generations including A100 and more
	Generative AI Studio - GUI for Models	Codey (Code)	Data (Human) Labelling	
	ColabEnterprise & Workbench Notebooks	ImageGen(Image)	FeatureStore	
	ModelGarden - Model Repository	Chirp (Translation) Dialogflow (conversations)	Google Marketplace offerings	
AWS	AWS Bedrock	AWS Titan	Amazon Redshift, Amazon RDS, Snowflake, Amazon Athena, and AWS Glue	AWS Trainium, AWS Inferentia, and NVIDIA GPUs.
	Huggingface	Vision & Recommendation		
	SageMaker (Build Models)	Amazon Lex for building chatbots.	Third party Foundation models - Cohere's & Claude 2 etc.	
	Amazon Comprehend - Find insights and relationships in text	Rekognition - Video & Image Third party Models	Vector engine for Amazon OpenSearch Serverless	
		Amazon Polly for text-to-speech,		
Azure	Azure OpenAI	GPT-4 models	Azure Machine Learning - Build, deploy, and manage models	ND H100 v5 Virtual Machine series, equipped with NVIDIA H100 Tensor Core graphics processing units (GPUs) and NVIDIA H100 GPUs interconnected by NVIDIA Quantum-2 InfiniBand networking.
	Azure AI Studio	Azure Bot Service for building chatbots	Microsoft Fabric - Unify, Manage & Govern Data	
	Copilots	DALL-E models		
By Ravi Saraswathi	Prompt Engineering	Azure Cognitive Services for computer vision	Azure App Services	

Source: https://www.linkedin.com/pulse/comparing-generative-ai-services-aws-azure-gcp-dr-rabi-prasad-3zqdc/

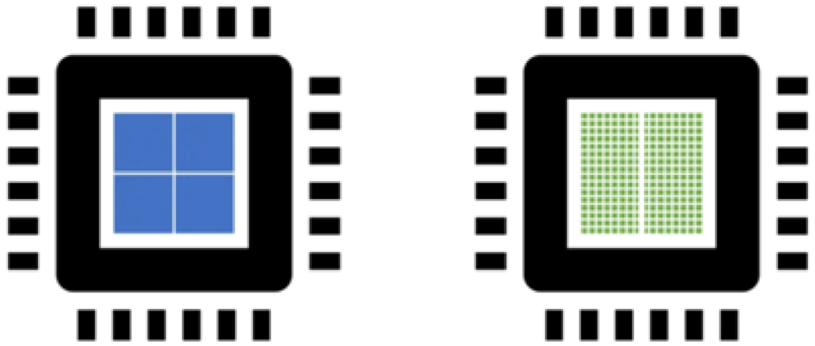
AWS GenAI Tools and Services



Source https://aws.amazon.com/blogs/awsmarketplace/generative-ai-partner-offerings-in-aws-marketplace-core-infrastructure-software/

Why GPUs Power GenAI

- GenAI training demands massive parallel processing of data from diverse sources, with thousands of simultaneous computations
- Traditional CPUs can't handle this scale, so GPUs step in
- GPUs (Graphics Processing Units) were originally built for rendering graphics but have evolved to perform complex mathematical operations
- Their ability to process many tasks in parallel makes them ideal for training large AI models efficiently

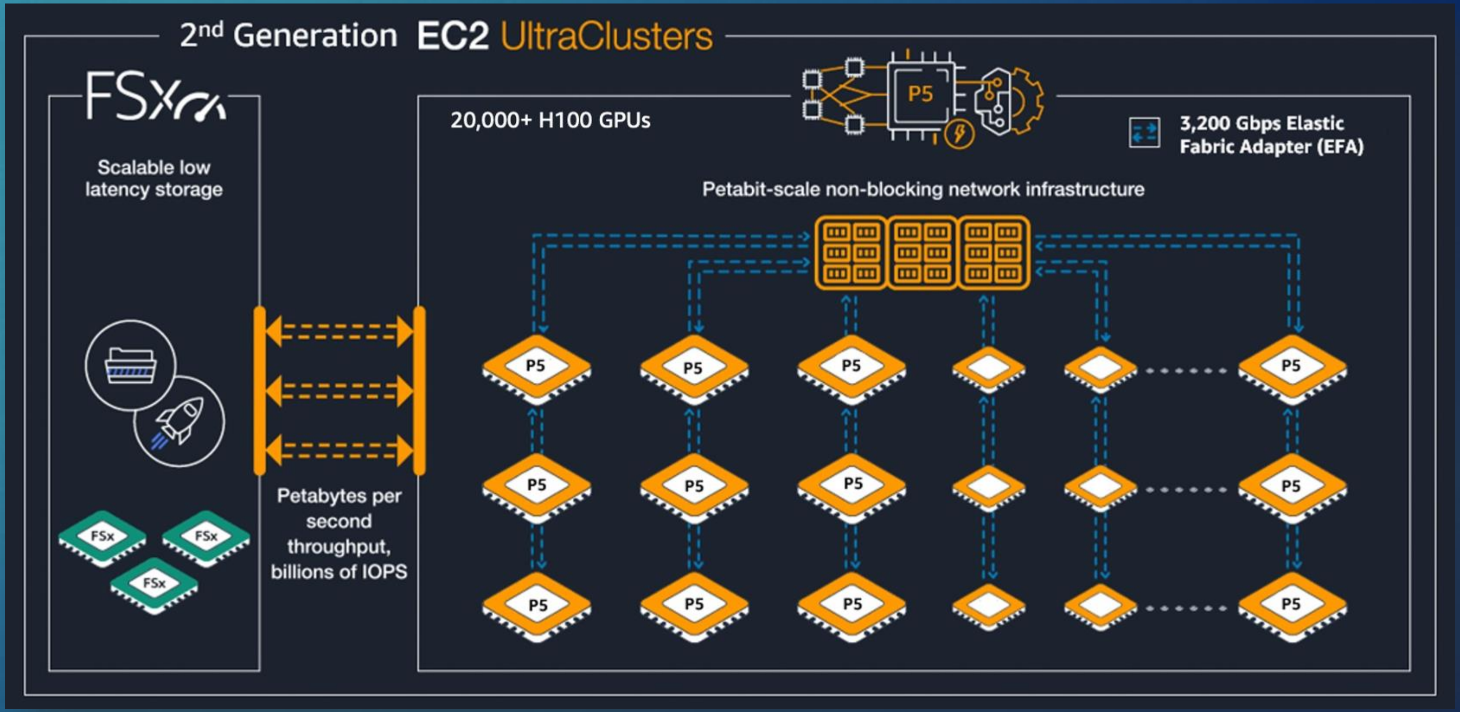


CPU	GPU
Central Processing Unit	Graphics Processing Unit
4-8 Cores	100s or 1000s of Cores
Low Latency	High Throughput
Good for Serial Processing	Good for Parallel Processing
Quickly Process Tasks That Require Interactivity	Breaks Jobs Into Separate Tasks To Process Simultaneously
Traditional Programming Are Written For CPU Sequential Execution	Requires Additional Software To Convert CPU Functions to GPU Functions for Parallel Execution

Source: https://www.cherryservers.com/blog/what-is-gpu-computing

AWS EC2 UltraClusters: High-Performance Infrastructure

- Thousands of accelerated EC2 instances co-located in an AWS Availability Zone and connected via Elastic Fabric Adapter (EFA) for ultra-fast networking.
- High-speed storage:
  - Local storage on P5 nodes
  - Access to FSx for Lustre, delivering petabytes per second throughput and billions of IOPS for massive parallel workloads



Source: https://aws.amazon.com/blogs/aws/new-amazon-ec2-p5-instances-powered-by-nvidia-h100-tensor-core-gpus-for-accelerating-generative-ai-and-hpc-applications/

AWS Bedrock

- Fully managed service for deploying and integrating generative AI models
- Provides API access to multiple foundation models without managing infrastructure
- Supports customization and fine-tuning for business-specific use cases
- Scales easily from small prototypes to enterprise workloads
- Key features:
  - Managed infrastructure to reduce operational overhead
  - Integration with AWS services (SageMaker, Lambda, and S3)
  - Access to models from Anthropic, Cohere, Meta, Stability AI, AI21 Labs, Mistral, and Amazon Titan
  - Enterprise-grade security with encryption and IAM controls

Amazon Bedrock

How does Amazon Bedrock work?

- Provides easy access to foundation models from leading providers via a single API
- Enables model switching or upgrades with minimal code changes
- Allows organizations adopt new GenAI innovations quickly without managing infrastructure or rewriting applications

Source: https://www.forbes.com/sites/moorinsights/2023/08/04/aws-turbocharges-foundation-models-with-smart-ai-agents/

Amazon Titan on AWS Bedrock

- Amazon Titan is a family of AWS-built **foundation models** available through Amazon Bedrock, designed for secure, scalable generative AI applications
- Titan models integrate natively with AWS services, enabling teams to build GenAI solutions without managing model infrastructure
- Titan model capabilities:
  - Text Premier – Generate and summarize text, answer questions, and power chat-based assistants
  - Image Generator – Create images from text prompts for use cases such as marketing, e-commerce, and media
  - Text Embeddings – Convert text into vectors for semantic search, recommendations, and similarity matching

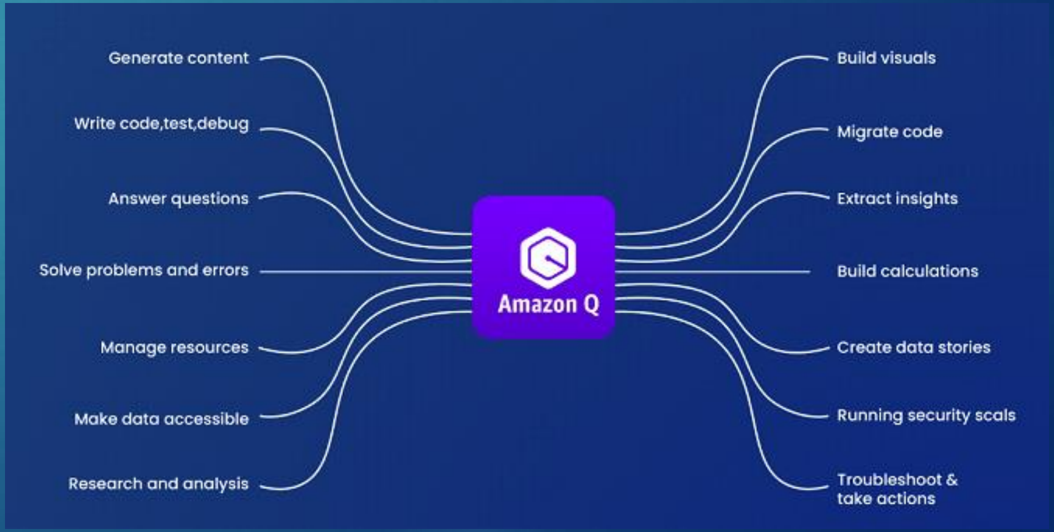
Source: https://blogs.idc.com/2024/07/31/embracing-the-future-how-genai-is-revolutionizing-cloud-infrastructure-for-growing-tech-vendors/

Amazon Q Developer

- Amazon Q is a managed, generative AI assistant built into AWS services to help users interact with cloud environments using natural language.
- Helps developers, operators, and business users ask questions and get answers about AWS resources and data
- Integrates directly with AWS services such as the AWS Console, documentation, and supported data sources
- Designed for enterprise use, with security, permissions, and governance built in
  - Offers a free model
  - Powered by Amazon Bedrock

Amazon Q Developer – How does it work?

- Users interact by asking questions in natural language
- Amazon Q gathers relevant context from AWS services, documentation, and approved data sources
- A fully managed foundation model uses that context to generate accurate responses
- Responses are permission-aware, honoring AWS IAM policies and access controls
- What you don't manage:
  - Model training
  - Infrastructure
  - Scaling or availability



GenAI Concerns and Ethical Challenges

Misinformation and Disinformation	• GenAI systems can generate responses that sound credible but are factually incorrect or misleading.
Bias and Fairness	• Models may reflect or amplify biases in training data, leading to unfair or discriminatory outcome
Privacy and Data Security	• Large training datasets may include sensitive data, raising concerns about data leakage or unintended disclosure
Intellectual Property and Copyright Infringement	• Generated content may resemble existing works, creating legal and ownership challenges.
Accountability and Transparency	• The "black box" nature of GenAI makes it difficult to explain decisions or assign responsibility
Human Oversight and Control	• Increased automation raises concerns about reduced human involvement in critical decisions
Environmental Impact	• Training and operating large models consumes significant compute and energy resources.

GenAI in Software Development - Today

- Code generation and refactoring (IDE copilots, API scaffolding)
- Test case creation and bug triage
- Documentation and knowledge base search
- Code review assistance and security scanning
- All enabled via cloud-hosted models and APIs (e.g., Bedrock, GitHub Copilot)

GenAI in Software Development – Future?

- From Coding to Orchestration
  - Developers will shift from writing code to designing and managing AI-driven workflows
- New Areas of Focus
  - Prompt Engineering – Crafting effective prompts for optimal AI output
  - Guardrails & Governance – Ensuring compliance, security, and ethical use
  - Cost & Performance Monitoring – Balancing efficiency with budget
- GenAI as a Built-In Foundation
  - AI becomes an integral part of cloud-native architectures, embedded by default
- Human Accountability Remains
  - Developers stay responsible for outcomes as AI accelerates work, not replaces it

Source: https://lab45thinktank.com/article/reimagining-software-engineering-with-genai-a-primer/

- Within your team, discuss the following:
  - Opportunities & Challenges
    - What excites you most about GenAI?
    - What concerns or risks do you see?
  - Tools & Innovations
    - Which GenAI tools or technologies have impressed you recently?
    - Any practical use cases you'd like to share?
  - Looking Ahead
    - Where do you think GenAI will have the biggest impact in the next 5–10 years?
    - How might it change the way we work?
- Teams have 10 minutes to share their responses